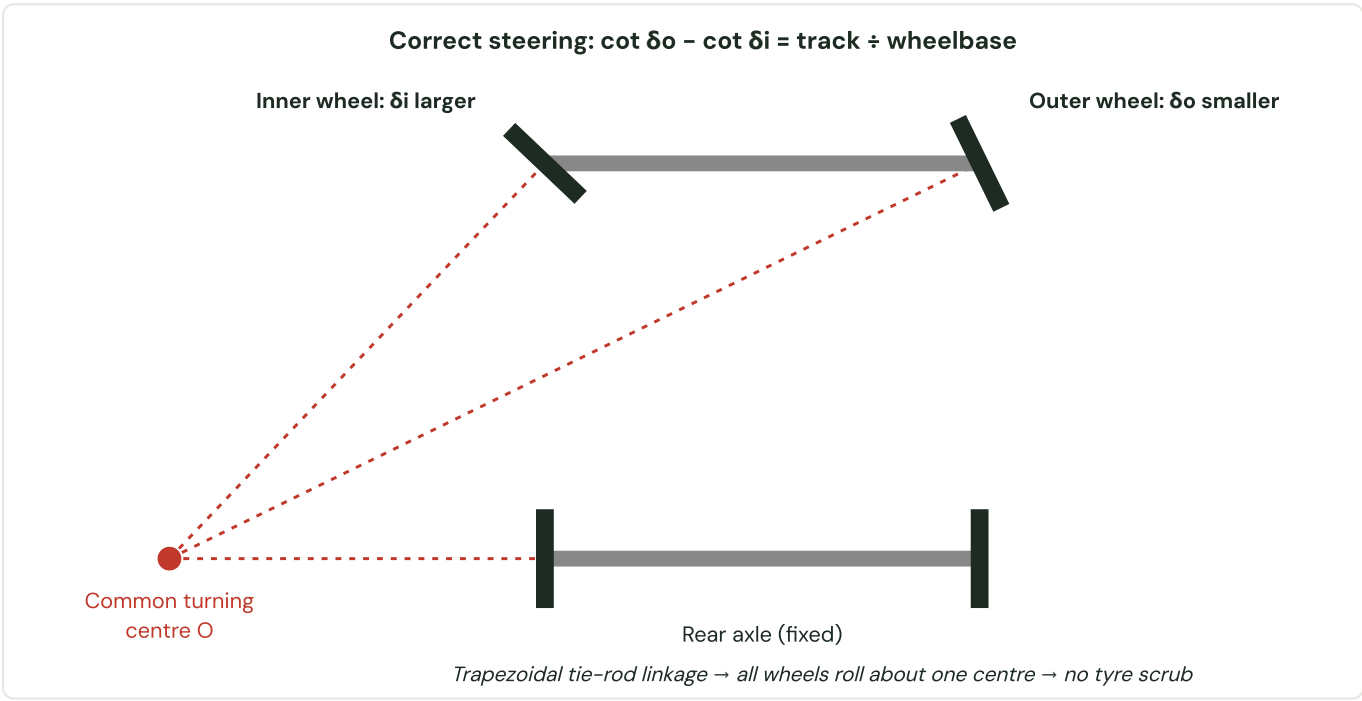
UNIT 4 · TOPIC 8

Mechanical and Power Steering

Syllabus topic: Mechanical and power steering

Mechanical steering linkage

1. Steering wheel.
2. Steering gear — worm & sector or recirculating ball (ratio ~**15–25:1**).
3. Drop (pitman) arm.
4. Drag link.
5. Steering arm.
6. Knuckles, coupled by the TIE ROD (trapezoidal Ackermann layout).



Ackermann steering geometry (redrawn from the source figure)

ACKERMANN condition: $\cot \delta_{outer} - \cot \delta_{inner} = \text{track/wheelbase}$

The INNER wheel steers through the LARGER angle, so all wheels roll about one turning centre on the rear-axle line — no tyre scrub.

Front-wheel geometry (definitions asked verbatim)

Term	Definition	Typical / purpose
TOE-IN	Front wheels closer at the front than at the rear	~3–12 mm — wheels run parallel under load
CAMBER	Outward tilt of the wheel top	~1–2°
CASTER	Backward tilt of the kingpin	~3° — self-centring
KINGPIN INCLINATION	Inward tilt of the kingpin	Aids return, reduces steering effort

Power steering

Hydraulic assistance, or full **HYDROSTATIC** steering (standard on current tractors):

1. The steering wheel drives a metering unit / orbital valve ("Orbitrol").
2. The valve directs pump oil to a steering cylinder on the axle.

- No mechanical shaft to the wheels; fewer linkage parts.
- Effortless steering with front loaders and on puddled fields.
- A manual back-up mode exists (harder).
- Electro-hydraulic GPS auto-steer taps the same circuit (Unit 10).

Practice MCQs

Q166. The correct-steering (Ackermann) condition is:

(a) cot δo + cot δi = B/L
(b) cot δo - cot δi = track/wheelbase
(c) tan δo = tan δi
(d) δo = δi

Q167. On a turn, the INNER front wheel is steered:

(a) Less than outer
(b) More than outer
(c) Same
(d) Not at all

Q168. Common tractor steering gears are:

(a) Rack only
(b) Worm & sector / recirculating ball
(c) Chain
(d) Bevel box

Q169. Toe-in of tractor front wheels is about:

(a) 0
(b) 3–12 mm
(c) 30–50 mm
(d) 100 mm

Q170. Outward tilt of the wheel top from vertical is:

(a) Caster
(b) Camber
(c) Toe-out
(d) KPI

Q171. Self-centring of steering after a turn is chiefly due to:

(a) Camber
(b) Caster (backward kingpin tilt)
(c) Toe-in
(d) Wide tyres

Q172. Full hydrostatic power steering uses:

(a) A drag link to wheels
(b) A metering (orbital) valve and steering cylinder with NO mechanical drive to wheels
(c) Chains
(d) Cable

Q173. Power steering is especially valuable with:

(a) Empty trailers
(b) Front-end loaders and puddled-field work
(c) Road transport only
(d) Small implements

Answers

Q166 — (b) cot δo - cot δi = track/wheelbase — *Geometry forces the inner wheel through a larger angle so both front wheels aim at the common centre on the rear-axle line.*

Q167 — (b) More than outer

Q168 — (b) Worm & sector / recirculating ball

Q169 — (b) 3–12 mm

Q170 — (b) Camber

Q171 — (b) Caster (backward kingpin tilt)

Q172 — (b) A metering (orbital) valve and steering cylinder with NO mechanical drive to wheels

Q173 — (b) Front-end loaders and puddled-field work

Sources: Liljedahl et al. — Tractors and their Power Units; Kepner et al. (hitching); IS 4468/ISO 730; Jagdishwar Sahay; NPTEL lectures.